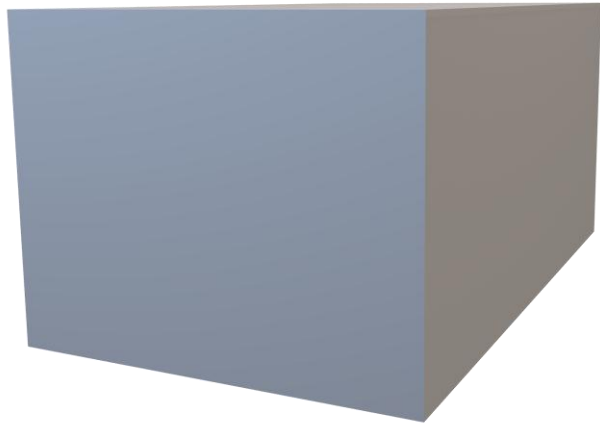


Automatic Refinement of 3D Building Models

LOD1.3 to LOD2.2 using Graph-Based Parametric Roof
Reconstruction with Machine Learning

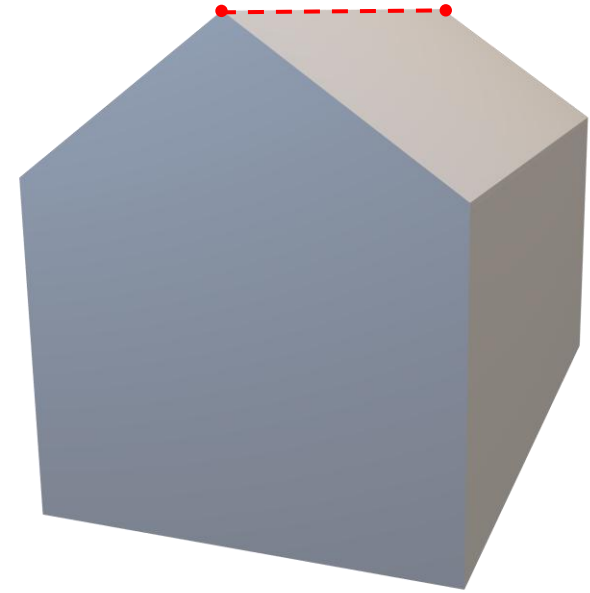
The Problem: Buildings without Roofs

Input Flat top (No roof shape)
LOD1.3



Pipeline

Output Actual roof geometry
LOD2.2



Research Questions

RQ1

How accurately can roof types (flat, gabled, hipped) be classified from LOD1.3 building geometry features using Random Forest classifier and which geometric features are most informative for this task?

RQ2

How accurately can roof type-specific geometric parameters (ridge height, ridge position) be predicted from LOD1.3 features using Random Forest regression, and what geometric accuracy does the resulting LOD2.2 reconstruction achieve compared to ground truth?

The Base Unit: What is a building in CityJSON?

CityJSON Structure

Building (parent)

Attributes only (no geometry)

b3_dak_type, b3_h_dak_min/max, volumes, etc

BuildingPart (child)

Contains the actual 3D geometry

LOD1.3 geometry (flat-top box)

LOD2.2 geometry (real roof shape)

Input = LOD1.3

Ground truth = LOD2.2

The LOD1.3 Solid

An extruded 2D footprint:

GroundFace – footprint polygon at ground level

TopFace – identical polygon at building height

WallFaces – Vertical rectangles connecting them

Typical rectangular building:

8 vertices (4 ground + 4 top)

6 faces (1 ground + 1 top + 4 walls)

No roof shape – just a flat top

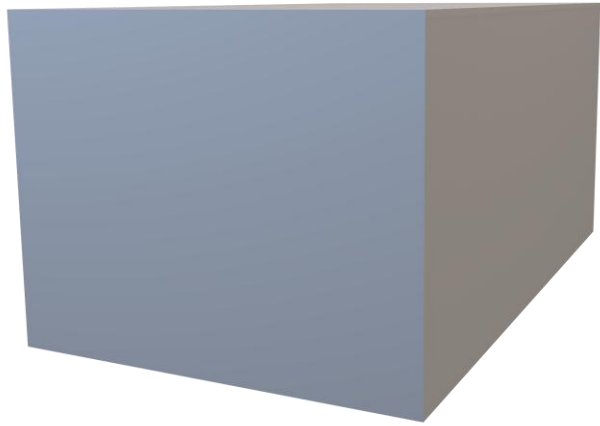
Coordinates: RD/NAP (EPSG:7415)

Stored as integers x 0.001 + translate

Three Roof Types We Handle

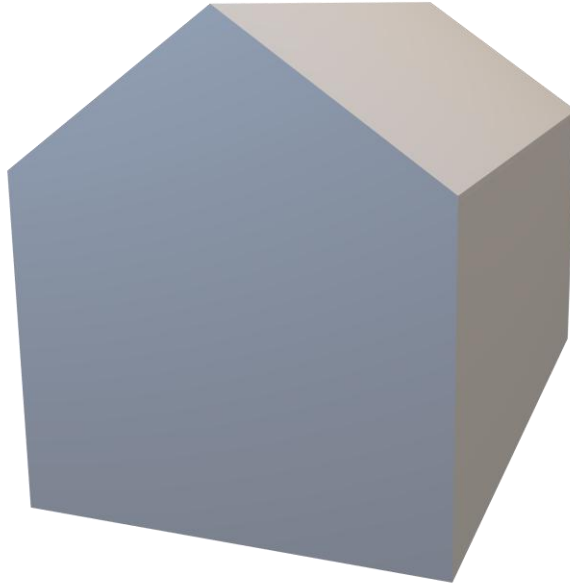
Flat

1 RoofSurface
0 Ridge Vertices



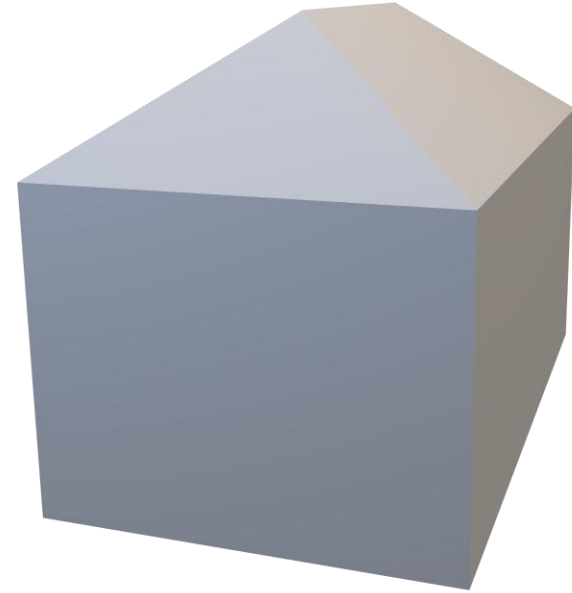
Gabled

2 RoofSurface
2 Ridge Vertices



Hipped

4 RoofSurface
2 Ridge Vertices



Roof type is determined by counting RoofSurface Faces in the LOD2.2 ground truth during training.

Dataset: 3DBAG

Source

Dutch 3D Building and Address dataset
10M+ buildings in CityJSON format
Both LOD1.3 and LOD2.2 provided
Publicly available at <https://3dbag.nl>
Coordinate system: EPSG:7415 (RD/NAP)

Subset

9 tiles processed(21,684 buildings)

8,542 buildings after filtering

Flat: 3993 buildings

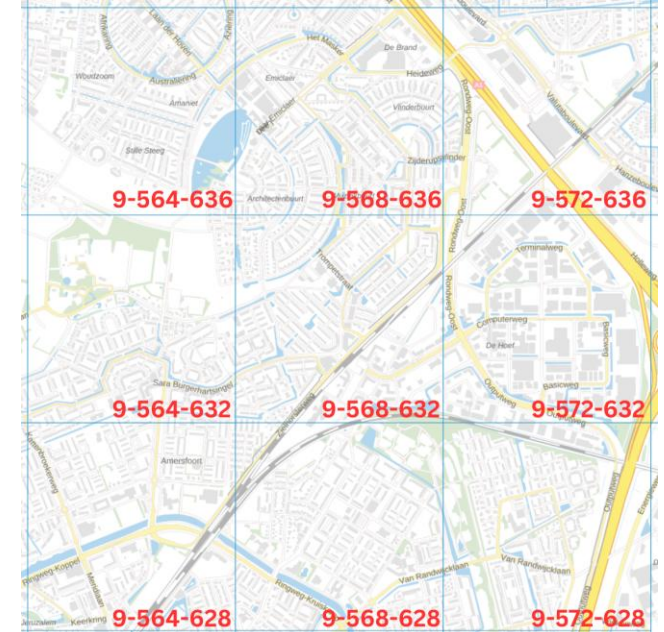
Gabled: 3624 buildings

Hipped: 925 buildings

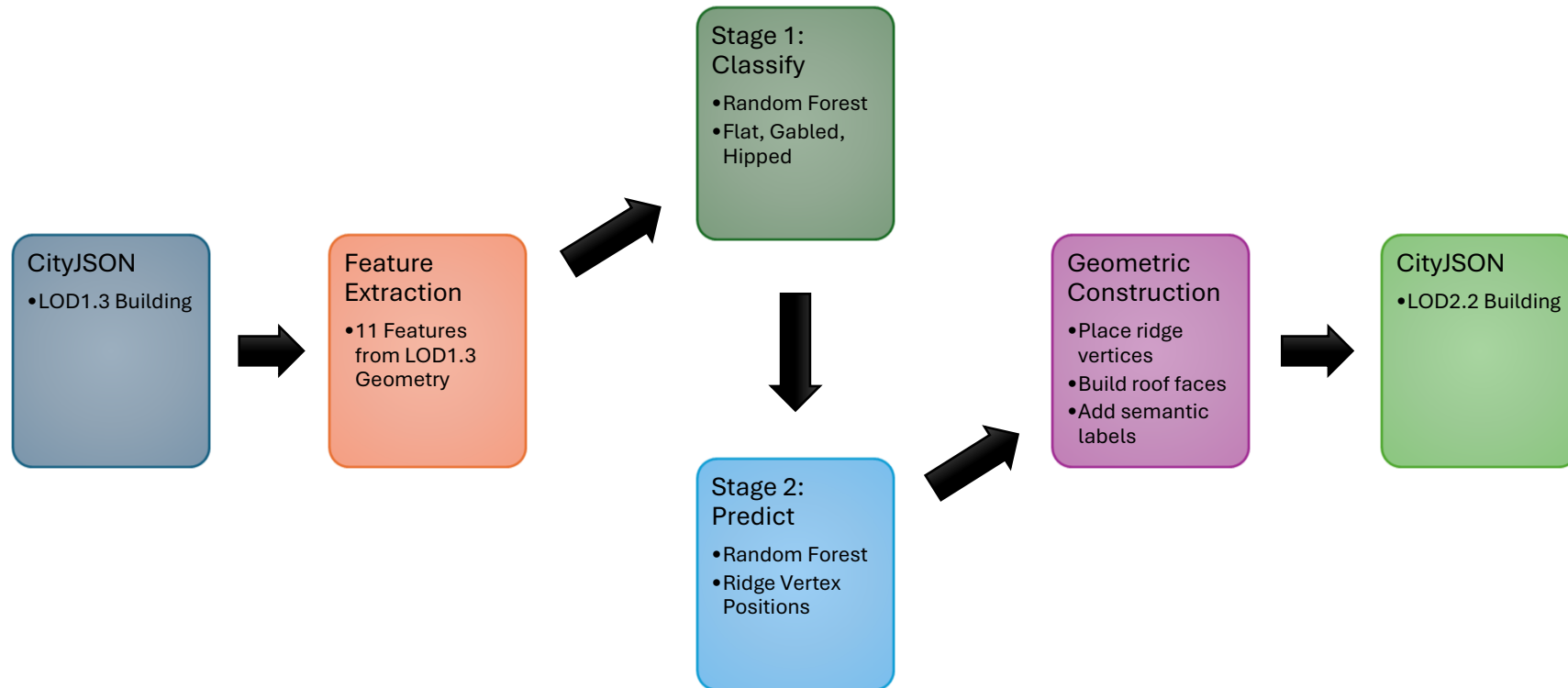
Filter:

Max 8 footprint vertices

Area 10-500 m²



Pipeline Overview



The pipeline has two ML stages: (1) classify the roof types, (2) predict the ridge vertex positions. Then deterministic geometric rules construct the final LoD2.2 building.

Feature Extraction and Selection

Started with 13 LOD1.3 features

Used 3 methods to select features:

- Random Forest Importance
- Permutation Importance
- Mutual Importance

Result: 13 → 7 features (accuracy: 88.8% → 88.6%)

Selected (7) – all from LOD1.3 geometry:

- building_height, vol_lod13, footprint_area, footprint_perimeter, mbr_length, mbr_width, longest_edge_length

Dropped (6):

- n_footprint_vertices, aspect_ratio, compactness, rectangularity, edge_length_ratio, orientation

Also excluded (LIDAR/LOD2.2-derived):

- h_dak_min/max, h_dak_50p/70p, height_range, height_std_proxy, vol_lod22, vol ratio, vol difference

Random Forest Feature Importance

During training, each tree splits data using features that best reduce prediction error.

Features that reduce error more often are considered more important.

So importance score reflects:

How much each feature reduces impurity across all trees?

Rank	Feature	Importance
1	building_height	0.2174
2	vol_lod13	0.1508
3	footprint_area	0.0963
4	footprint_perimeter	0.0833
5	mbr_width	0.0810
6	mbr_length	0.0761
7	longest_edge_length	0.0584
8	aspect_ratio	0.0527
9	compactness	0.0475
10	edge_length_ratio	0.0432
11	rectangularity	0.0382
12	n_footprint_vertices	0.0311
13	orientation	0.0241

Permutation Importance

Measures importance after the model is trained.

Procedure:

1. Train model normally
2. Shuffle one feature randomly
3. Measure how much accuracy drops

If accuracy drops a lot → feature is important.

If accuracy barely changes → feature is not useful.

Meaning of Standard Deviation is importance was measured multiple times, and results varied slightly.

Small std = stable importance estimate.

Rank	Feature	Importance	Std
1	building_height	0.2606	0.0090
2	longest_edge_length	0.0276	0.0033
3	vol_lod13	0.0218	0.0045
4	mbr_length	0.0140	0.0035
5	footprint_perimeter	0.0126	0.0036
6	mbr_width	0.0101	0.0024
7	rectangularity	0.0094	0.0029
8	n_footprint_vertices	0.0085	0.0034
9	aspect_ratio	0.0073	0.0027
10	footprint_area	0.0072	0.0019
11	orientation	0.0071	0.0016
12	edge_length_ratio	0.0068	0.0022
13	compactness	0.0061	0.0023

Mutual Importance

Mutual Information measures how much information a feature contains about the target variable.

$MI(X,Y)$ = how much knowing X reduces uncertainty about Y

High MI → strong relationship. Low MI → weak relationship.

Rank	Feature	MI Score
1	building_height	0.4776
2	vol_lod13	0.3614
3	mbr_length	0.3585
4	footprint_perimeter	0.3436
5	longest_edge_length	0.3384
6	mbr_width	0.3367
7	footprint_area	0.3311
8	orientation	0.2103
9	edge_length_ratio	0.1739
10	compactness	0.1279
11	aspect_ratio	0.1192
12	rectangularity	0.0614
13	n_footprint_vertices	0.0559

Consensus Feature Selection

Selected features (7):

footprint_area	score=3 (RF rank:3, Perm rank:10, MI rank:7)
footprint_perimeter	score=6 (RF rank:4, Perm rank:5, MI rank:4)
mbr_length	score=6 (RF rank:6, Perm rank:4, MI rank:3)
mbr_width	score=6 (RF rank:5, Perm rank:6, MI rank:6)
building_height	score=6 (RF rank:1, Perm rank:1, MI rank:1)
longest_edge_length	score=5 (RF rank:7, Perm rank:2, MI rank:5)
vol_lod13	score=6 (RF rank:2, Perm rank:3, MI rank:2)

Dropped features (6):

n_footprint_vertices	score=2 (RF rank:12, Perm rank:8, MI rank:13)
aspect_ratio	score=2 (RF rank:8, Perm rank:9, MI rank:11)
compactness	score=2 (RF rank:9, Perm rank:13, MI rank:10)
rectangularity	score=2 (RF rank:11, Perm rank:7, MI rank:12)
edge_length_ratio	score=2 (RF rank:10, Perm rank:12, MI rank:9)
orientation	score=1 (RF rank:13, Perm rank:11, MI rank:8)

Key Findings (RQ1)

Top feature: building_height (21.7%) followed by vol_lod13 (15.1%) and footprint_area (9.6%)
Building size and height alone provide strong classification signal from pure LOD1.3 geometry
7 features sufficient all derivable from LOD 1.3 without LIDAR or LOD2.2 data

Roof Type Classification

Input (7 features from LOD1.3)

- footprint_area (m²)
- footprint_perimeter (m)
- mbr_length (longest side of bounding rect)
- mbr_width (shortest side of bounding rect)
- longest_edge_length (m)
- building_height (max_z – min_z)
- vol_lod13 (LOD1.3 volume in m²)

Output (Roof types)

One of three classes:

Flat | Gabled | Hipped

RF Accuracy: 0.8900

RF F1 Score: 0.8920

Predicted → Actual ↓	flat	gabled	hipped
flat	560	27	12
gabled	32	466	46
hipped	1	23	115

	precision	recall	f1-score	support
flat	0.94	0.93	0.94	599
gabled	0.90	0.86	0.88	544
hipped	0.66	0.83	0.74	139

MLP Accuracy: 0.8619

MLP F1: 0.8597

Roof Type Classification

But how can roof type be predicted using only footprint geometry?

In architecture, the roof structure is strongly influenced by the footprint geometry.

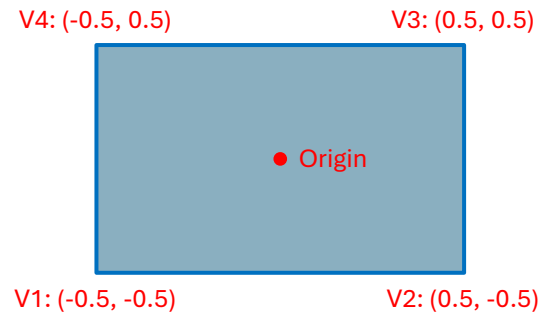
Reasons include:

- structural stability (wall placement, span length, load distribution)
- Drainage (helps direct water away from walls)
- construction cost
- architectural design patterns (warehouse → flat, suburban house → gabled, compact house → hipped)

So certain footprint shapes tend to use certain roof types.

Roof Vertex Prediction

Input

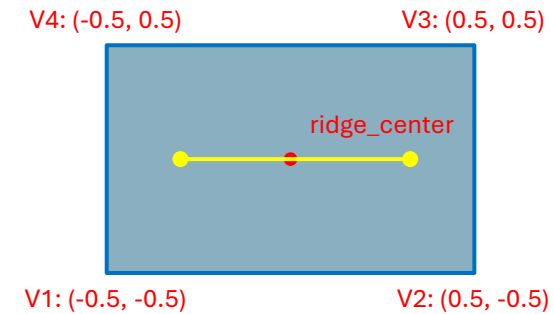


Each building is represented using 26 geometric features derived from the footprint and building properties.

Input dimension: 26

Separate model for gabled and hipped roof.

Output



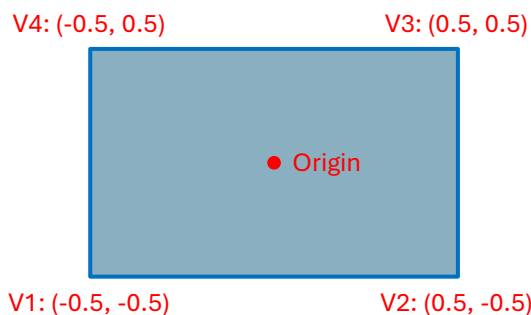
Output Vector:

[ridge_cx, ridge_cy, ridge_z, ridge_half_len, dim4, dim5]

Instead of predicting the entire roof mesh, the model predicts ridge parameters that allow reconstruction of the roof.

Roof Vertex Prediction

Input Representation



Footprint shape:

Normalised coordinates of up to 8 footprint vertices.

[v1_x, v1_y, v2_x, v2_y, ..., v8_x, v8_y]

Footprint is centered at building centroid

Coordinates scaled by max(MBR length, width)

footprints with fewer vertices are zero-padded

Geometric Building Features:

len(footprint_2d), normalized_area, aspect_ratio,
compactness, rectangularity, normalized_height,
edge_length_ratio, normalized_orientation,
normalized_mbr_length, normalized_mbr_width

Models Tested:

Random forest

MLP-small: 64 → 32

MLP-medium: 128 → 64 → 32

MLP-large: 256 → 128 → 64

Gabled Roof:

Model	MAE
MLP-small	0.0518
MLP-medium	0.0563
MLP-large	0.0520
Random Forest	0.0343

Hipped roof:

Model	MAE
MLP-small	0.0894
MLP-medium	0.0899
MLP-large	0.0730
Random Forest	0.0427